Chapter 1

MULTI-FREQUENCY APPROACH

In this chapter, the procedure for identifying flutter derivatives using both the Single-Frequency and Multi-Frequency methods from CFD simulations is introduced. The calculation of the critical wind speed using the aerodynamic and structural properties of the bridge deck is then presented. A Python tool is developed to perform these tasks and validated against benchmark results.

This chapter is organised as follows:

1. Identification of Aerodynamic Forces
2. Calculation of Flutter Derivatives from Aerodynamic Forces
3. Calculation of Critical Wind Speed from Flutter Derivatives
4. Multi-Frequency Approach

1.1 Identification of Aerodynamic Forces

Aerodynamic forces are obtained using forced-motion CFD simulations, as described below.

- A 2-D cross-section of the bridge deck is modelled in a virtual wind tunnel with appropriate boundary conditions.

- The bridge cross-section is oscillated sinusoidally in the vertical (heave) direction:

$$h(t) = h_0 \sin(\omega t), \tag{1.1}$$

$$\dot{h}(t) = \omega h_0 \cos(\omega t), \tag{1.2}$$

  where h_0 is the maximum amplitude of the applied heave motion and ω is the angular frequency of the applied heave motion.

- Since no pitch motion is applied, $\alpha = \dot{\alpha} = 0$. The aerodynamic lift force and pitching moment therefore simplify to

$$L_{ae} = \frac{1}{2}\rho U^2 B \left[K H_1^* \frac{\dot{h}}{U} + K^2 H_4^* \frac{h}{B} \right] \qquad \text{(Scanlan formulation)}, \tag{1.3}$$

$$M_{ae} = \frac{1}{2}\rho U^2 B^2 \left[K A_1^* \frac{\dot{h}}{U} + K^2 A_4^* \frac{h}{B} \right] \qquad \text{(Scanlan formulation)}. \tag{1.4}$$

- Under the small-amplitude assumption (small h_0), Scanlan's linearisation implies that the aerodynamic lift and moment measured from CFD take the sinusoidal form

$$L_{ae} = B_0 \sin(\omega t + \phi) \;=\; B_1 \cos(\omega t) + B_2 \sin(\omega t), \tag{1.5}$$

$$M_{ae} = C_0 \sin(\omega t + \Phi) \;=\; C_1 \cos(\omega t) + C_2 \sin(\omega t). \tag{1.6}$$

1.2 Calculation of Flutter Derivatives

- Equating the lift force from the Scanlan formulation with the CFD result and matching cosine and sine coefficients gives

$$\frac{1}{2}\rho U^2 B \left[K H_1^* + K^2 H_4^*\right] = B_1 \cos(\omega t) + B_2 \sin(\omega t), \tag{1.7}$$

$$\frac{1}{2}\rho U^2 B^2 \left[K A_1^* + K^2 A_4^*\right] = C_1 \cos(\omega t) + C_2 \sin(\omega t). \tag{1.8}$$

  Hence the flutter derivatives for heave excitation are

$$H_1^* = \frac{2B_1}{\rho\,\omega^2 B^2 h_0}, \qquad H_4^* = \frac{2B_2}{\rho\,\omega^2 B^2 h_0}, \tag{1.9}$$

$$A_1^* = \frac{2C_1}{\rho\,\omega^2 B^3 h_0}, \qquad A_4^* = \frac{2C_2}{\rho\,\omega^2 B^3 h_0}. \tag{1.10}$$

- The same procedure is repeated with pitch (torsional) excitation to obtain the remaining flutter derivatives H_2^*, H_3^*, A_2^*, and A_3^*.

- In total, two forced-motion simulations (one heave, one pitch) are required to obtain a complete set of eight flutter derivatives at each reduced velocity.

1.3 Calculation of Critical Wind Speed

Equations of Motion

The coupled two-degree-of-freedom equations of motion for the bridge deck are

$$m\Big(\ddot{h} + 2\omega_h \zeta_h \dot{h} + \omega_h^2 h\Big) = L_{ae}, \tag{1.11}$$

$$I_\alpha\big(\ddot{\alpha} + 2\omega_\alpha \zeta_\alpha \dot{\alpha} + \omega_\alpha^2 \alpha\big) = M_{ae}, \tag{1.12}$$

where the aerodynamic forces are given by the Scanlan formulation:

$$L_{ae} = \frac{1}{2}\rho U^2 B \left[K H_1^* \frac{\dot{h}}{U} + K H_2^* \frac{B\dot{\alpha}}{U} + K^2 H_3^* \alpha + K^2 H_4^* \frac{h}{B}\right], \tag{1.13}$$

$$M_{ae} = \frac{1}{2}\rho U^2 B^2 \left[K A_1^* \frac{\dot{h}}{U} + K A_2^* \frac{B\dot{\alpha}}{U} + K^2 A_3^* \alpha + K^2 A_4^* \frac{h}{B}\right]. \tag{1.14}$$

Matrix Form (Time Domain)

Substituting the aerodynamic forces into the equations of motion and rearranging yields the matrix system

$$\begin{aligned}&\begin{bmatrix} m & 0 \\ 0 & I \end{bmatrix} \begin{Bmatrix} \ddot{h} \\ \ddot{\alpha} \end{Bmatrix} + \begin{bmatrix} C_h & 0 \\ 0 & C_\alpha \end{bmatrix} \begin{Bmatrix} \dot{h} \\ \dot{\alpha} \end{Bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\alpha \end{bmatrix} \begin{Bmatrix} h \\ \alpha \end{Bmatrix} \\ &- \frac{1}{2}\rho K^2 \left(\begin{bmatrix} H_1^* & BH_2^* \\ A_1^* & BA_2^* \end{bmatrix} \begin{Bmatrix} \dot{h} \\ \dot{\alpha} \end{Bmatrix} + \begin{bmatrix} H_4^* & H_3^* \\ A_4^* & BA_3^* \end{bmatrix} \begin{Bmatrix} h \\ \alpha \end{Bmatrix} \right) = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}.\end{aligned} \tag{1.15}$$

Frequency-Domain Eigenvalue Problem

Assuming harmonic motion $h = e^{i\omega t}$ and $\alpha = e^{i\omega t}$, and transforming to the frequency domain, the system becomes

$$\left(-\omega^2 \begin{bmatrix} m & 0 \\ 0 & I \end{bmatrix} + i\omega \begin{bmatrix} C_h & 0 \\ 0 & C_\alpha \end{bmatrix} + \begin{bmatrix} K_h & 0 \\ 0 & K_\alpha \end{bmatrix} - \frac{1}{2}\rho K^2 \begin{bmatrix} H_4^* - H_1^* & BH_3^* - BH_2^* \\ A_4^* - A_1^* & BA_3^* - BA_2^* \end{bmatrix} \right) \begin{Bmatrix} h \\ \alpha \end{Bmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \tag{1.16}$$

which is of the standard form

$$-\omega^2 M + i\omega C + K = 0. \tag{1.17}$$

Here $\omega = \omega_r + i\omega_i$ is a complex frequency: the real part ω_r is the oscillation frequency and the imaginary part ω_i governs the growth or decay of the response.

Stability Criterion

Because structural damping is present the problem is a complex quadratic eigenvalue problem, which is linearised as follows:

$$\left(\begin{bmatrix} 0 & K \\ K & iC \end{bmatrix} - \omega \begin{bmatrix} K & 0 \\ 0 & -M \end{bmatrix} \right) \begin{pmatrix} X \\ \omega X \end{pmatrix} = \begin{Bmatrix} 0 \\ 0 \end{Bmatrix}, \qquad \text{i.e.} \quad A - \lambda B = 0. \tag{1.18}$$

The sign of ω_i determines stability:

- $\omega_i > 0$: $h = h_0\, e^{i\omega_r t} \cdot e^{-\omega_i t}$ — **stable** (amplitude decays).
- $\omega_i < 0$: $h = h_0\, e^{i\omega_r t} \cdot e^{+|\omega_i| t}$ — **unstable** (amplitude grows; flutter onset).

A Python tool has been developed to solve this linearised eigenvalue problem and identify the critical wind speed.

Determining the critical wind speed is an iterative process:

1. A reduced velocity $U_r = U/(fB)$ is selected.
2. Flutter derivatives corresponding to that reduced velocity are interpolated from the flutter-derivative functions.
3. The linearised eigenvalue problem is solved for ω.
4. If the system is stable, the reduced velocity is increased and the process repeats until the system first becomes unstable.
5. The actual critical wind speed is then recovered from

$$U_c = U_{r,c} \frac{\omega_r}{2\pi} B, \tag{1.19}$$

where $U_{r,c}$ is the reduced critical velocity and ω_r is the real part of the eigenfrequency at that point.

1.4 Validation of the Critical Velocity Tool

The Python tool is validated against the benchmark results of Diana [**?**].

Structural Parameters

Table 1.1 lists the structural properties used in the validation, taken from Diana [**?**].

Table 1.1: Structural parameters used for validation (Diana benchmark).

Symbol	**Description**	**Value**
m_L	Mass per unit length	22 740 kg/m
J_L	Moment of inertia per unit length	2.47×10^6 kg m^2/m
B	Deck chord (width)	31 m
f_z	Vertical structural eigenfrequency	0.10 Hz
f_θ	Torsional structural eigenfrequency	0.278 Hz
ξ	Damping ratio (both modes)	0.003 (−)

Aerodynamic Parameters

In the benchmark, Theodorsen derivatives are used for the aerodynamic properties of the cross-section. The Theodorsen flutter derivatives are analytical functions derived from the closed-form solution for a thin flat plate in potential flow, first obtained by Theodorsen [**?**]. They are expressed in terms of the real and imaginary parts of the Theodorsen circulatory function $C(k) = F(k) + iG(k)$, where k is the reduced frequency. The analytical expressions are listed in Table 1.2 and plotted in Figure 1.1.

Table 1.2: Analytical Theodorsen flutter derivative expressions for a flat plate.

	H_i^*	A_i^*
$i = 1$	$H_1^* = -\dfrac{2\pi}{K} F$	$A_1^* = \dfrac{\pi}{2K} F$
$i = 2$	$H_2^* = -\dfrac{\pi}{2K}\left(1 + F + \dfrac{4G}{K}\right)$	$A_2^* = -\dfrac{\pi}{8K}\left(1 - F - \dfrac{4G}{K}\right)$
$i = 3$	$H_3^* = -\dfrac{2\pi}{K^2}\left(F - \dfrac{KG}{4}\right)$	$A_3^* = \dfrac{\pi}{2K^2}\left(\dfrac{K^2}{32} + F - \dfrac{KG}{4}\right)$
$i = 4$	$H_4^* = \dfrac{\pi}{2}\left(1 + \dfrac{4G}{K}\right)$	$A_4^* = -\dfrac{\pi}{2K} G$

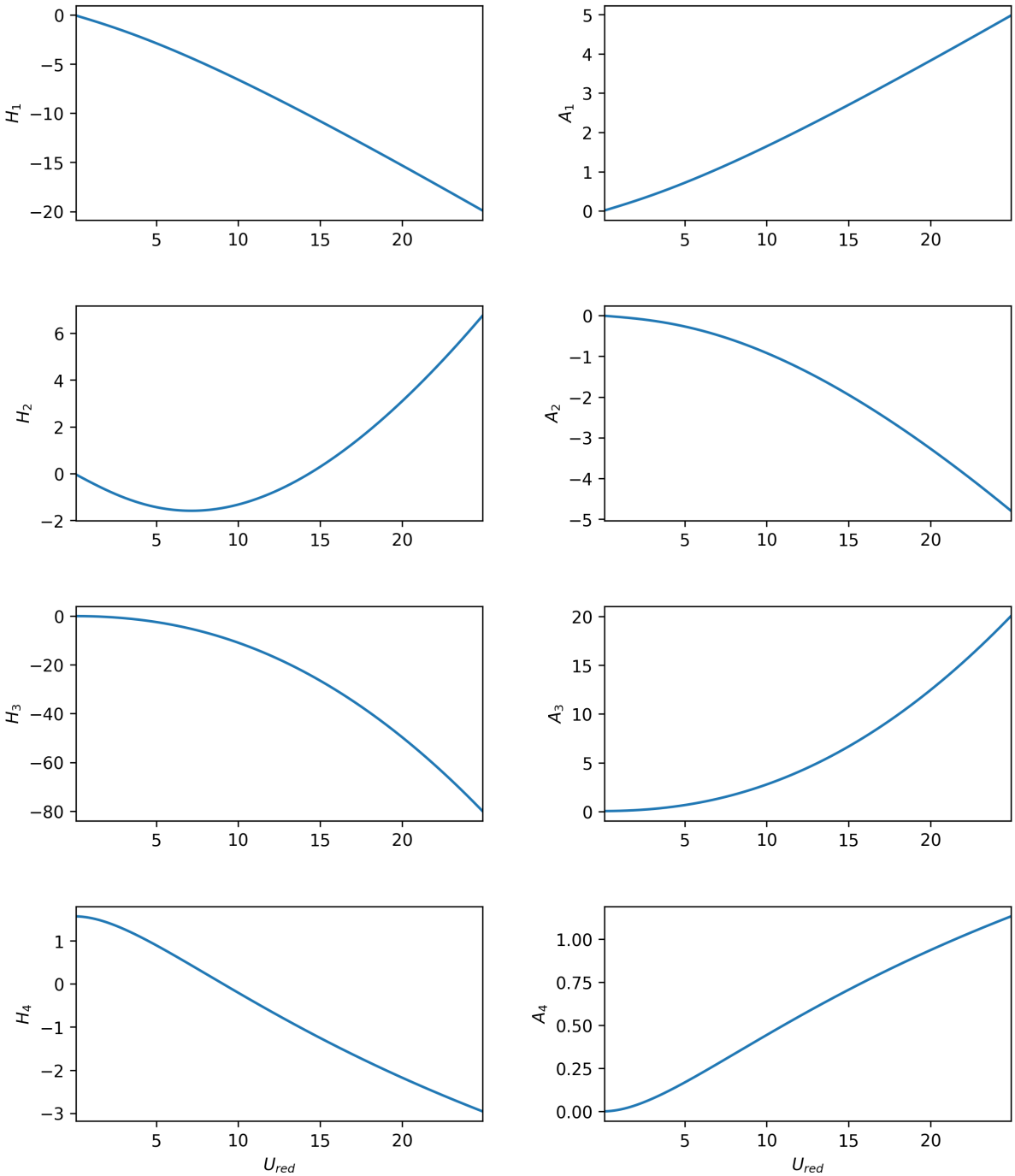

Figure 1.1: Theodorsen Flutter Derivatives

Results

The results obtained with the Python tool are shown in Figures 1.2 and 1.3.

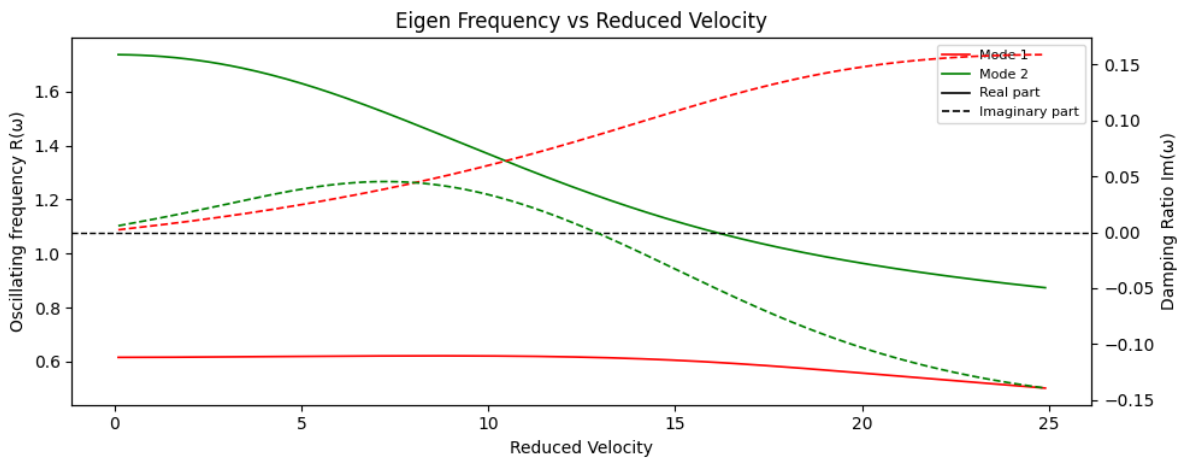

Figure 1.2: Real and imaginary parts of the eigenfrequency plotted against reduced velocity

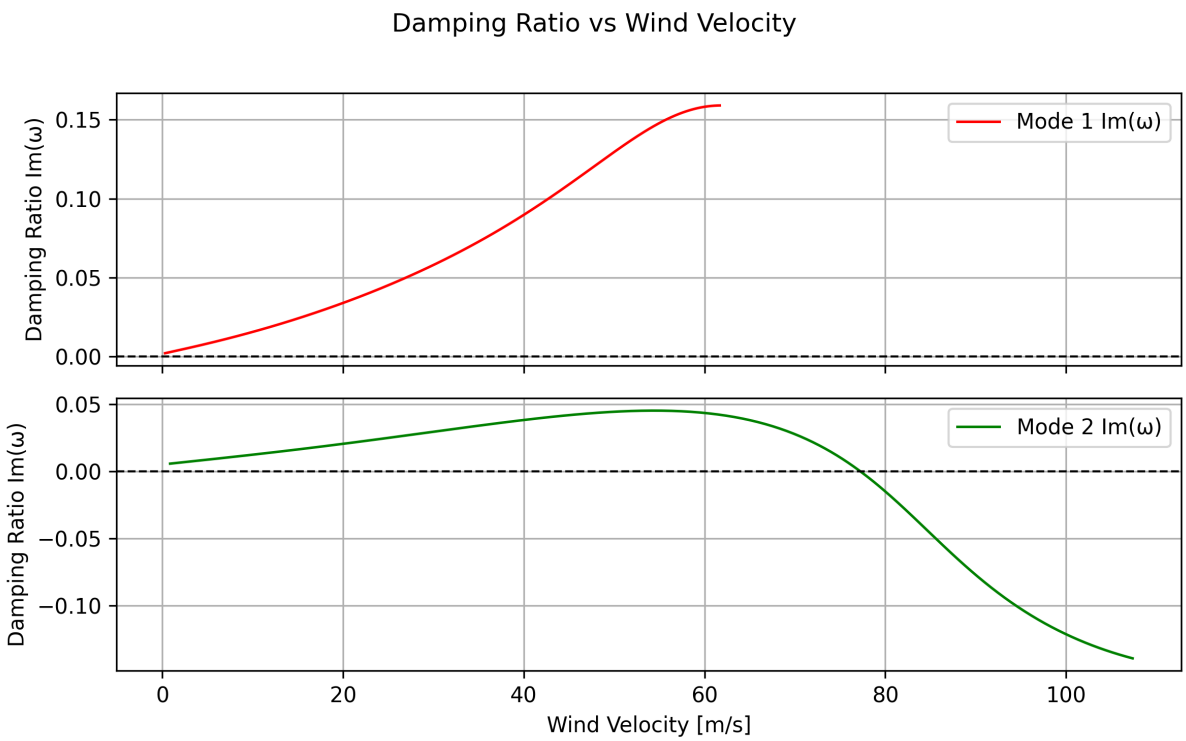

Figure 1.3: Damping ratio plotted against wind velocity.

Since the bridge deck is modelled as a two-degree-of-freedom system, two eigenmodes are obtained. The zero-crossing of the imaginary part of each eigenfrequency is checked; the lowest reduced velocity at which a zero crossing occurs is taken as the reduced critical velocity $U_{r,c}$. Substituting into Equation (1.19) yields

$$
\begin{aligned}
U_{r,c} &= 12.9, \\
\omega_r &= 1.21 \text{ rad/s}, \\
U_c &= 77.32 \text{ m/s}.
\end{aligned}
$$

This result falls within the reference range of [77.1, 78.6] m/s reported by Diana [**?**], confirming the validity of the tool.

1.5 Multi-Frequency Approach

In the single-frequency approach, the forced-motion oscillation (heave or pitch) is applied at one frequency at a time, yielding one set of flutter derivatives per simulation. In the **multi-frequency approach**, several frequencies are superimposed in a single forced-motion simulation:

$$
h(t) = \sum_{k=1}^{N} h_{0,k} \sin(\omega_k t + \phi_k). \tag{1.20}
$$

Under the assumption of aerodynamic linearity, the total aerodynamic response is the superposition of the responses at each individual frequency. Consequently, N sets of flutter derivatives can be extracted from a single simulation by applying a Fourier decomposition of the measured aerodynamic forces at each frequency component ω_k.

The procedures for calculating the flutter derivatives (Section 2) and the critical wind speed (Section 3) are identical for both the single- and multi-frequency approaches. The only difference lies in the identification of aerodynamic forces (Section 1), where multiple frequency components are excited simultaneously rather than one at a time. Figures 1.4 and 1.5 illustrate the workflow for the two approaches.

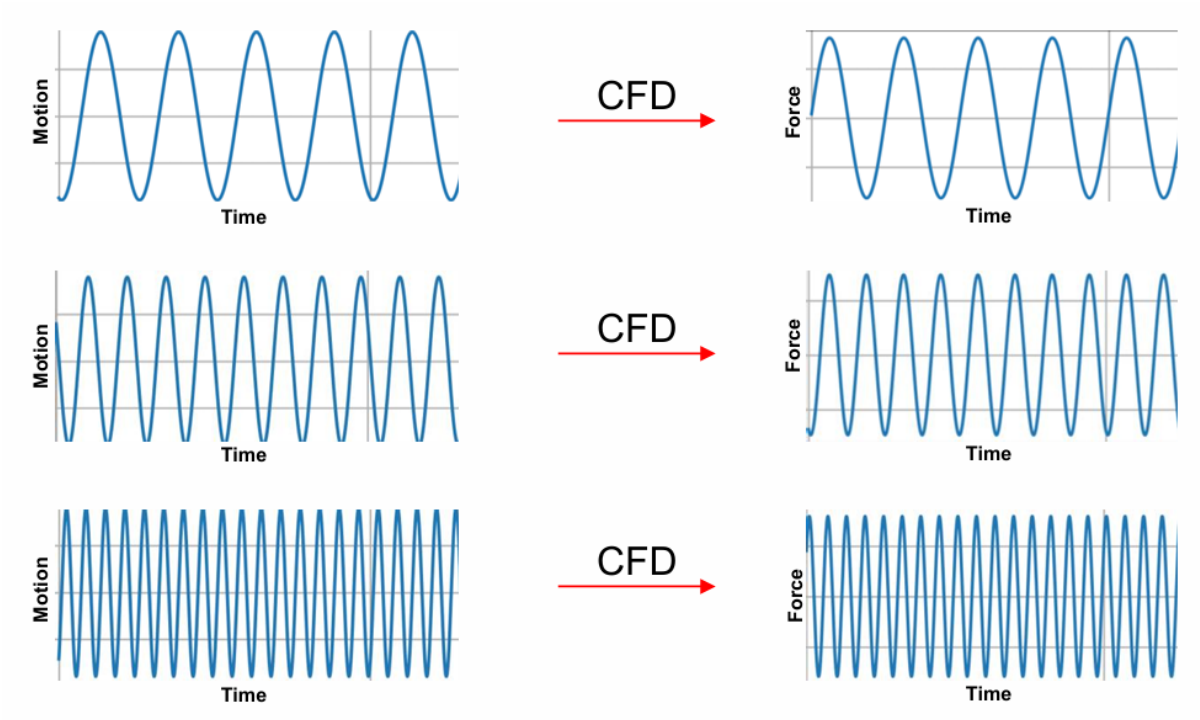

Figure 1.4: Workflow for the single-frequency approach.

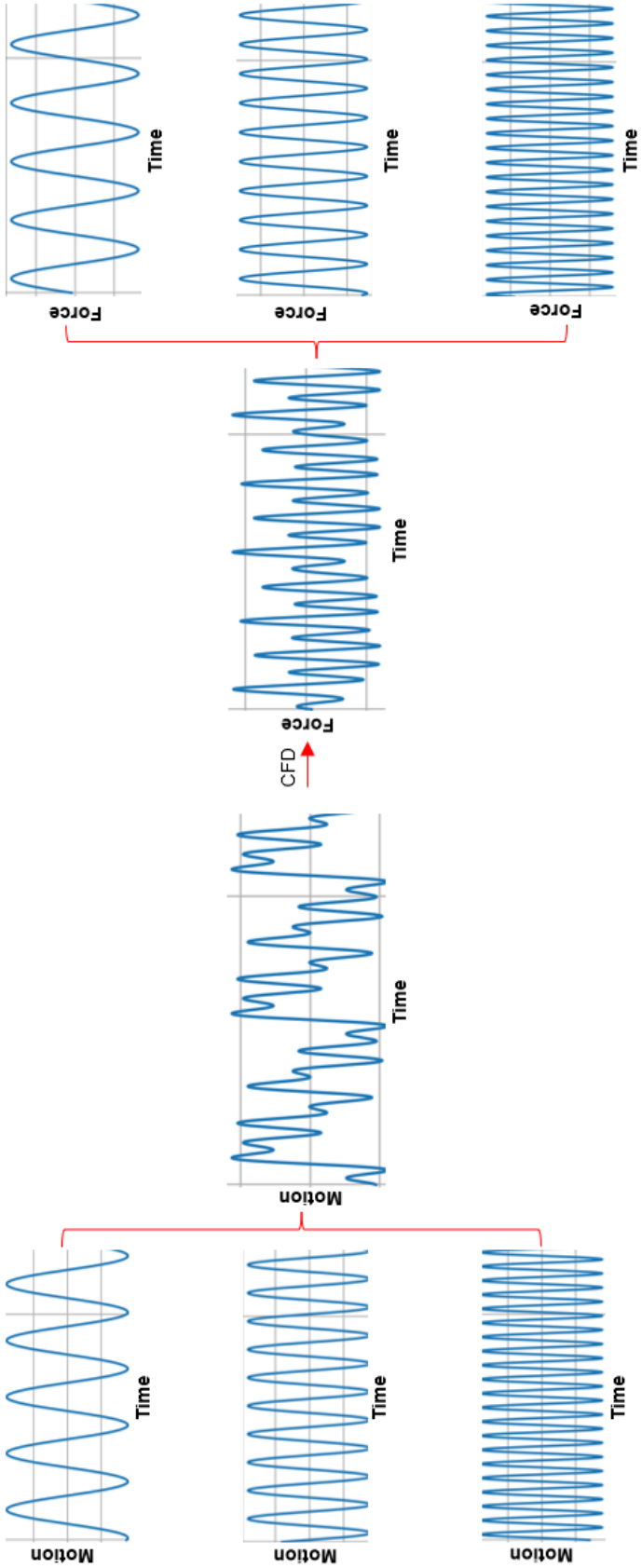

Figure 1.5: Workflow for the multi-frequency approach.